REGRESSION PROJECT

MATH 564 (Fall 2025) Under Prof. Maggie Cheng

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This project aims to apply the tools learned in class to analyze the Linthrust data and identify the important physiochemical properties of the substrate influencing aerial biomass production in the Cape Fear Estuary of North Carolina. We have the response variable Y_BIO, which is the biomass production, and 14 predictor variables characterizing the soil.

All the analysis given below is based on the values obtained after running the codes. For this project, I have used MATLAB. Readers are advised to run the codes themselves in order to get the values and make their own observations. However, all the figures obtained for this project have been attached.

ANALYSIS FOR PART I OF THE PROJECT:

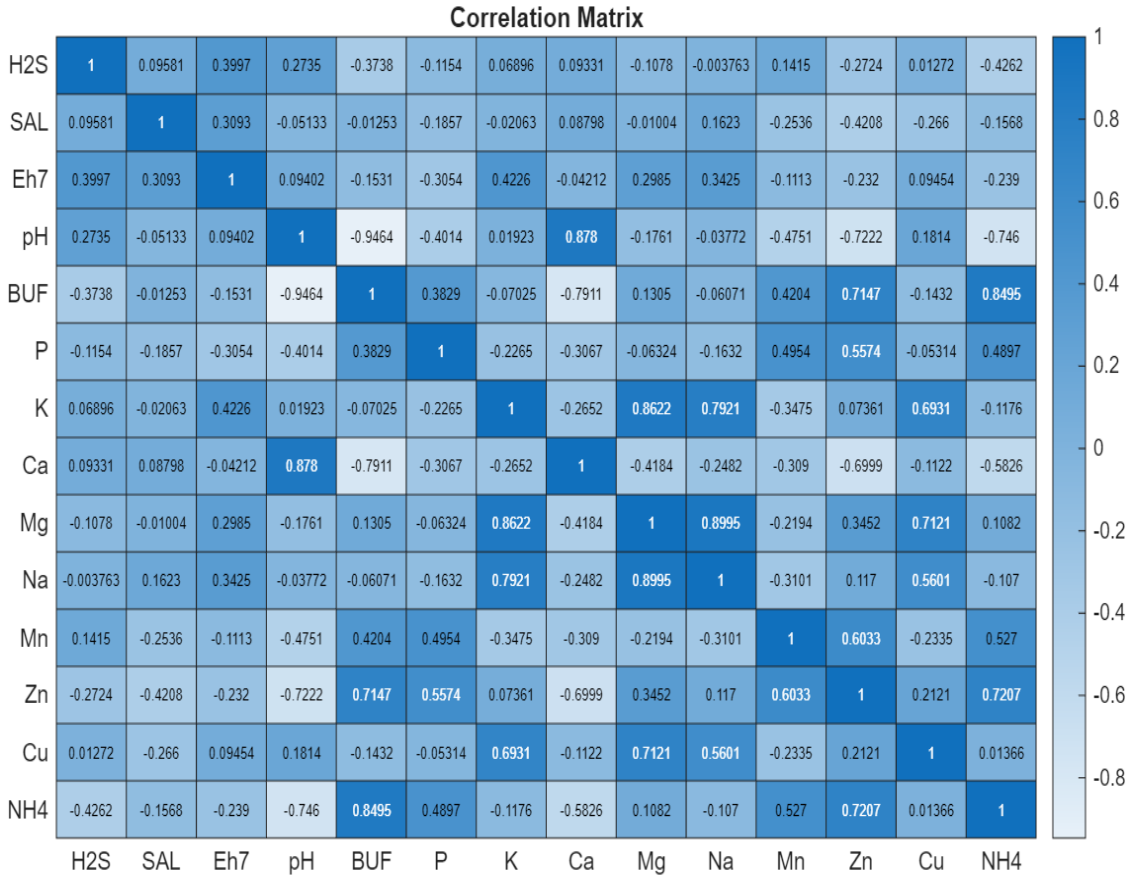
For the three collinearity diagnostic methods that I applied to the data set LINTHALL, given below are my observations.

- VARIANCE INFLATION FACTOR: Many predictors had extremely large VIF values (>10), which shows severe multicollinearity. The predictors were pH, BUF, Ca, Mg, Na, Zn. This shows that many predictors are linearly dependent or almost dependent. Only H2S, SAL, Eh7, P had VIF <4, which is ideal for us. The values that I obtained are listed below.

H2S	SAL	Eh7	pH	BUF	P	K
3.0275	3.3876	1.9774	62.081	34.432	1.8958	7.3671

Ca	Mg	Na	Mn	Zn	Cu	NH4
16.662	23.764	10.351	6.1856	11.626	4.8292	8.3765

- CORRELATION MATRIX: Many predictor pairs had high pairwise correlations, such as Ca-Mg, Na-Mg, pH-BUF, to name a few. These strong correlations confirm redundancy among predictors. Refer to the heatmap for an overall idea.



- CONDITION INDICES/EIGENVALUE DIAGNOSTICS: The condition indices were also pretty large, with some values being between 10 and 30, which gives us possible moderate multicollinearity. It means that small changes in the data can cause large changes in the coefficients. This shows that the design matrix X is nearly singular. The condition indices obtained are:

  22.7752
  19.7300
  13.0116
  11.8586
  8.5356
  7.2617
  6.3721
  6.3334
  4.1814
  3.8812
  3.0251
  2.1843
  1.7734
  1.0000.

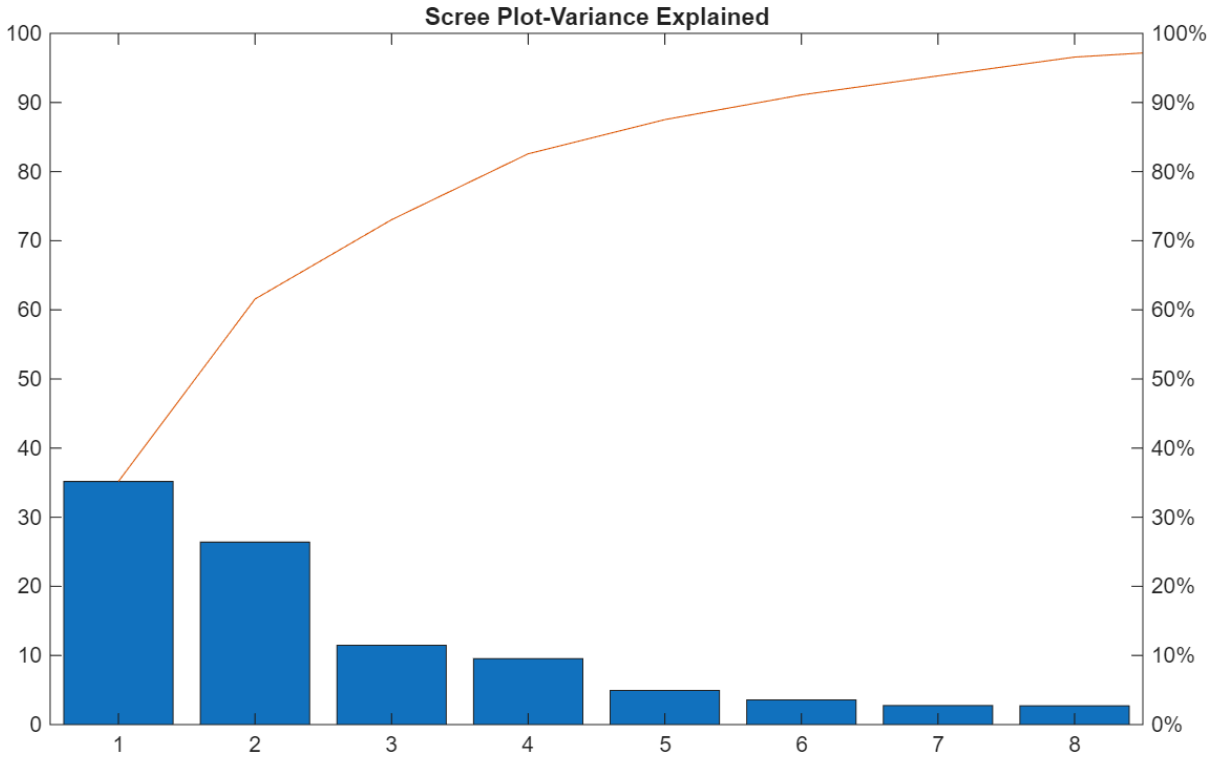
  (The order is the same as above.)

Next, I performed Principal Components Regression and computed the regression coefficients in the original multiple linear regression model. After standardizing the predictors, I performed principal component analysis.

The first few principal components captured the majority of the total variation in the predictors. The leading components with substantial eigenvalues were retained. These principal components explained most of the variation while removing noise and instability from the model. I used the criteria of cumulative variance being greater than 95 for the predictors. The coefficients obtained after a regression model was fit using selected principal components were transformed back into the original predictor scale. Given below are the results.

H2S	**SAL**	**Eh7**	**pH**	**BUF**	**P**	**K**	**Ca**	**Mg**
129.69	-94.061	-2.0945	137.39	-71.704	-41.328	-5.4801	66.235	-101.88

Na	**Mn**	**Zn**	**Cu**	**NH4**
-176.24	-99.672	-107.61	154.36	-17.73

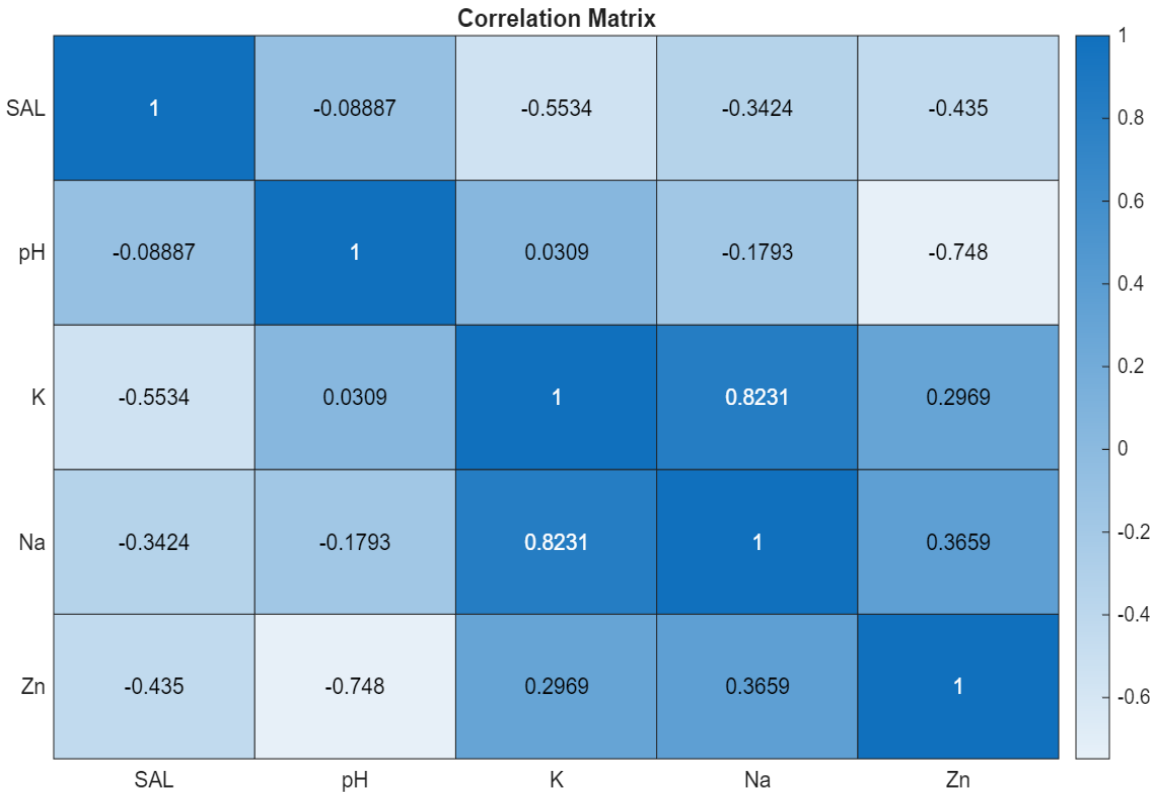


Based on the results, we can infer that PCR produced much more stable coefficients since the principal components are orthogonal, as compared to OLS coefficients, which were highly unstable due to multicollinearity. The sum of standard errors in OLS was very large, sum of standard errors (OLS)=4048.0882, which was substantially reduced by PCR, sum of standard errors (PCR)=500.3847, confirming a more stable estimation. Also, based on the SSE obtained for both PCR and OLS, one can say that PCR provides a more efficient representation of data without overfitting.

ANALYSIS FOR PART II OF THE PROJECT:

For this part of the project, the data set considered is LINTH-5. This data set has 5 predictor variables, which preserves some of the collinearity problem. The following observation is based on the results obtained after performing a variable selection procedure on this data. It can be observed that the collinearity here is moderate, as the variance inflation factor does not exceed 6 for any of the predictors. The values obtained are:

SAL	**pH**	**K**	**Na**	**Zn**
2.7769	4.4804	4.5378	3.7642	5.4386



For the stepwise regression, starting with a constant model, I considered each variable for entry based on the significant partial F-tests. After adding and removing variables, the final model given by the algorithm is
Linear regression model:
BIO ~ 1 + pH + Zn,

With estimated coefficients:

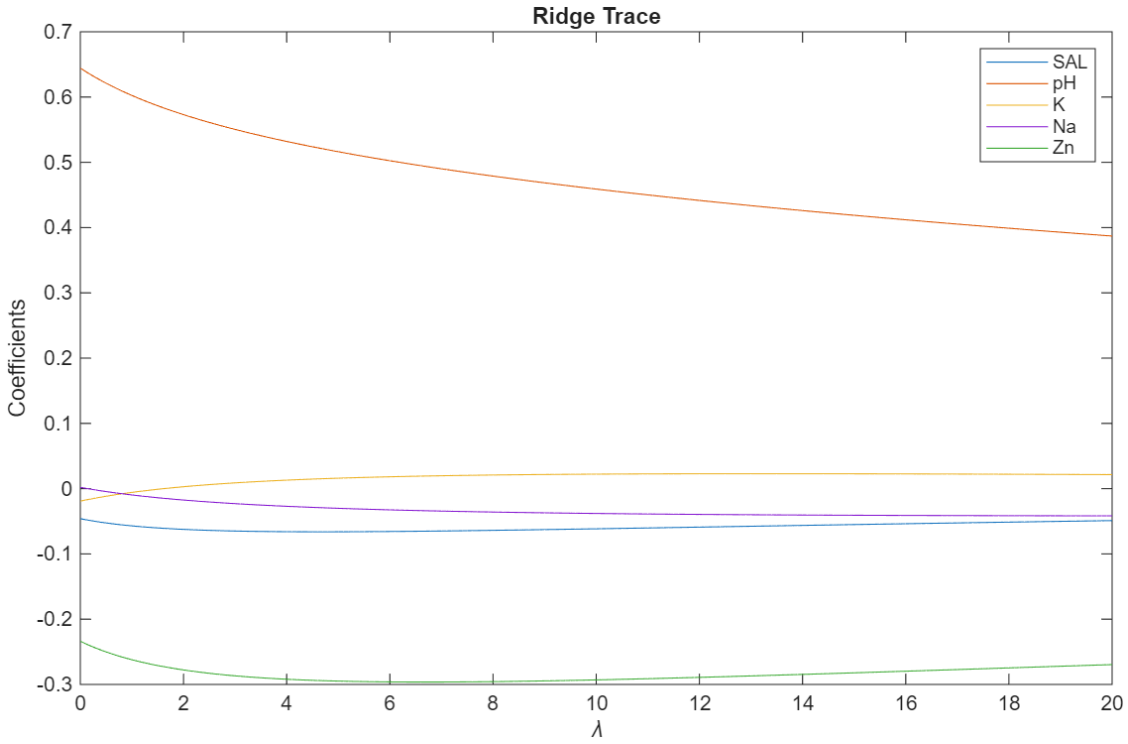
	Estimate	**SE**	**tStat**	**pValue**
(Intercept)	-250.05	524.02	-0.47717	0.63639
pH	350.25	75.21	4.657	5.0497e-05
Zn	-15.236	11.404	-1.3361	0.19067

Stepwise regression selected the predictors that were statistically significant after accounting for the other variables. The final model was smaller than the full model due to high multicollinearity among the original predictors.

Further, for the best subset selection, all possible two-predictor models were fit using the five available predictors. The best subset procedure consistently favored models that contained SAL plus one other variable. This also aligns with the strong collinearity among the data. The values obtained for AIC, BIC, and SSE, with the lowest values highlighted are given below for different combinations.

comb		AIC	BIC	SSE
1	2	88.761	93.511	21.001
1	3	482.63	487.38	1.1851e+06
1	4	703.53	708.28	5.4786e+08
1	5	225.73	230.48	943.12
2	3	495.33	500.08	1.6864e+06
2	4	708.14	712.89	6.2275e+08
2	5	234.79	239.54	1213
3	4	665.72	670.47	1.9165e+08
3	5	236.42	241.17	1269.3
4	5	237.91	242.66	1322.9

At last, I applied Ridge regression to standardized predictors over a range of λ values from 0 to 20. The plot below shows how the coefficient estimates changed as λ increased.



From the Ridge trace, the strongest and most stable predictors were SAL and pH, which I kept in my code. Using these variables, a reduced ridge model was fit, and variance inflation factor dropped, coefficients became more stable, and the model avoided the collinearity problem seen in OLS.

Across both datasets, all three collinearity diagnostics (VIFs, correlation matrices, and condition indices) consistently showed that the predictors were highly multicollinear. Because of this, the full OLS models were unstable and difficult to interpret.
Dimension-reduction and model-selection methods provided more reliable alternatives: PCR in Part 1 produced stable coefficients and reduced variance, while stepwise regression, best subset selection, and ridge regression in Part 2 all favored smaller models built from only the strongest predictors.
Overall, the results show that reduced models outperform the full models in both parts, and effective prediction of BIO can be achieved using only a few key variables once multicollinearity is addressed.